Predicting Coronary Heart Disease from the BRFSS Survey

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***Abstract*—We predict coronary heart disease (MICHD) from the 2015 Behavioral Risk Factor Surveillance System (BRFSS), a telephone survey of over 300,000 US adults with 321 coded answers each. Only 8.8% of respondents are positive, so we optimize F1 and tune the decision threshold. We first build regularized logistic regression trained with Newton's method as a baseline (test F1 0.439). We then decode the survey using its codebook, recovering units, skip-logic answers and missingness patterns, and train our own gradient-boosted decision trees in NumPy, which reach a cross-validated F1 of 0.4435 ± 0.0038 and a test F1 of 0.451.**

I. Introduction

Coronary heart disease is a leading cause of death [1], and many of its risk factors (age, hypertension, smoking, diabetes, inactivity) are captured by self-reported lifestyle surveys. Our task is to predict whether a BRFSS respondent has ever been diagnosed with MICHD, i.e. heart attack or angina.

The data is hard for three reasons. (i) It is a raw survey, not a clean feature matrix: answers mix units (e.g. weight in pounds or kilograms), packed encodings ("1xx" per week, "2xx" per month), and special codes for "don't know" or "refused". (ii) Many questions are only asked after a filter question, so most missing values carry information instead of being random. (iii) The classes are imbalanced (8.8% positive), which makes accuracy misleading and F1 sensitive to the decision threshold.

We describe our feature processing in Sec. II, a regularized logistic regression baseline in Sec. III, and our gradient-boosted trees in Sec. IV, and compare them in Sec. V.

II. Feature Processing

The final GBDT pipeline uses codebook-based preprocessing [2]; the logistic baseline uses the separate processing described in Section III. Identifiers are excluded from predictors and used to align observations with labels.

We map feature-specific unknown and refused responses to NaN, encode binary answers as 1/0, and convert packed frequencies to common units. Missing follow-ups are filled when a preceding response determines their value: for example, no alcohol consumption implies zero drinking quantities. Other missing values remain NaN.

We merge household adult counts across telephone questionnaires and derive diabetes duration, an indicator of HIV testing within five years, and a cognitive-impact score averaging the available responses across three reversed response scales. `UNSURE_CHRONIC` indicates an explicit "don't know" response to any of eleven chronic-condition questions.

We remove administrative variables, survey weights, selected optional-module questions, and redundant representations, retaining 176 features. The final tree pipeline applies no additional percentile capping or imputation. Histogram boundaries are fitted on training observations and reused for test data, with a separate bin for missing values. Nominal variables are marked for categorical splitting. `HAREHAB1` is excluded from predictors and used only in the override described in Section V.

III. Baseline: Regularized Logistic Regression

Our first model minimizes the logistic loss with an L_2 penalty, $\frac{1}{N}\sum_i \ell(y_i, \mathbf{x}_i^\top \mathbf{w}) + \lambda\|\mathbf{w}\|^2$. Gradient descent needed hundreds of steps to converge, so we use Newton's method: each step solves $(\frac{1}{N}X^\top SX + 2\lambda I)\,\Delta\mathbf{w} = \nabla L$, with $S = \mathrm{diag}(p_i(1-p_i))$, and the loss stops changing after about six steps. A linear model cannot handle missing values or unordered categories by itself, so we impute missing values with the training median, add a binary "was missing" indicator per feature, one-hot encode categorical columns, add squared terms for continuous columns, and standardize all features. We choose $\lambda \in \{10^{-4}, 10^{-3}, 10^{-2}\}$ and the decision threshold jointly by 5-fold cross-validation on F1. This model reaches a test F1 of 0.439 (Table II).

IV. Gradient-Boosted Decision Trees

The idea of GBDT [3] is to build an additive model on the log odds, $s(\mathbf{x}) = s_0 + \eta\sum_{k=1}^{K} f_k(\mathbf{x})$, where $f_k(\mathbf{x})$ represents k-th tree, and s_0 represents base rate of the positive class, that is:

$$s_0 = \log\frac{\hat{p}}{1-\hat{p}}, \quad \hat{p} = \frac{1}{n}\sum_i \mathbf{1}\{y_i = 1\}.$$

The goal of the method is to have a sequence of trees that would sequentially improve the scores. At each round, we compute for every single sample the gradient $g = p - y$, where $p = \sigma(s(\mathbf{x}))$ and the Hessian $h = p(1-p)$, and fit one regression tree to all of them. The intuition is as follows: Suppose that we have $y = 1$ and current prediction probability $p = 0.3$, it's clear to see that the gradient will be given by $g = -0.7$ which means that we should increase

the score. Similarly, the Hessian represents the curvature, or the confidence in our prediction, as we can see that the expression will be maximized as $p \to \frac{1}{2}$, and is minimized for confident predictions $p \to \{0, 1\}$.

At each leaf we take a Newton-step $-\frac{G_\ell}{H_\ell + \lambda_{\text{leaf}}}$, where G_ℓ and H_ℓ are the sums of g and h over that leaf. This is the result of minimizing the second-order Taylor expansion over parameter w of the expression:

$$\sum_i \ell(y_i, s_i + w) \approx \text{const} + G\, w + \tfrac{1}{2}(H + \lambda)\, w^2,$$

where $G = \sum g_i,\ H = \sum h_i$ [4].

Each tree outputs w_{leaf} for all the samples in it. We can see that splitting a node into left and right children gives us the following advantage:

$$\text{gain} = \tfrac{1}{2}\left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{G^2}{H + \lambda}\right],$$

where L and R indicate left and right child, respectively. We only split a node if both children have $H_L, H_R \geq H_{min}$, so that we do not create trees for small number of already confident samples.

Finally, since finding the splitting thresholds would be too slow if we took into account all the possible thresholds, we decide on binning our values inside 64 bins, where bin 0 corresponds to NaN values, while the other 63 bins are computed on the train set and are ordered. For ordered values, the bins smaller than the threshold go to the left child and the larger ones to the right, while the 0-th bin can go either way. For categorical bins, we use the trick from [5] so that we overcome combinatorial nature of all possible permutations.

We fit each tree on 70% random rows and columns (features), with learning rate $\eta = 0.05$, $H_{min} = 10$, and regularization parameter $\lambda_{\text{leaf}} = 1$. All randomness is seeded, so the submission is exactly reproducible.

A. Results

To test the GBDT model we use 5-fold CV. To determine the number of trees we use early stopping with patience 50 epochs, on 10% hold-out of the current training fold, which also serves to choose the F1-maximizing decision threshold. Since this hold-out contains only ≈2,300 positives, the threshold is noisy (0.20–0.25 across folds); for the final model we therefore also choose it on the pooled out-of-fold predictions of the other four folds, which never uses the scored fold. Table I summarizes the results.

For features, we started with 97 processed features which were first used in the linear model, where GBDT achieves F1 of 0.4403, already greater than all logistic variants we previously tried. Additionally, giving it all 176 recoded columns from preprocessing with NaNs kept (NaNs are in the previously mentioned 0-th bin.) increases F1 by $+0.002$ and lowers average log-loss $0.2190 \to 0.2181$. With the threshold from the other folds, the best value is $\tau = 0.21$ in every fold and we achieve F1 of 0.4435 ± 0.0038 across 5 folds using τ.

Table I
GBDT RESULTS. TOP: 5-FOLD CV (MEAN) BOTTOM: FOLD 0, LOG-LOSS $\times 10^3$.

5-fold CV	F1	Log-loss
97 imputed features	0.4403	0.2190
176 recoded, NaN kept	0.4423	0.2181
+ threshold $\tau = 0.21$	**0.4435**	0.2181
Fold 0	$\eta = 0.1$	$\eta = 0.05$
depth 3	216.85	–
depth 4	216.69	**216.24**
depth 5	216.63	216.32
depth 4 + dropped raw cols.	216.62–217.34	–

Table II
TEST-SET (AICROWD) RESULTS. THE ALL-NEGATIVE ACCURACY IS ESTIMATED FROM THE TRAINING POSITIVE RATE.

Model	F1	Accuracy
Always predict -1	0	≈0.91
Reg. logistic regression (Newton)	0.439	0.876
GBDT + `HAREHAB1` override	**0.451**	**0.877**

Regarding hyperparameters, we do small search on fold 0. We find that depth 3 is too shallow for different interactions, and depths 4/5 are very close to each other (these results hold for learning rate $\eta = 0.1$). Lowering $\eta = 0.05$, changes the picture such that now depth 4 achieves the lowest log-loss, and for this depth the folds stop at 464 trees on average.

For this model (depth 4, $\eta = 0.1$) adding raw leftover columns decreases log-loss by no more than 10^{-4}, with F1 difference around ± 0.002, which are not always followed by the sign of log-loss difference.

The final model trained on the full training set is constructed with 500 trees (to account for an increase in data) and achieves **F1 0.451** and **Accuracy 0.877**, for threshold $\tau = 0.21$.

V. COMPARISON

On the hidden test set, GBDT improves F1 over our best regularized logistic regression from 0.439 to 0.451 (Table II). Accuracy stays essentially unchanged (0.876 vs. 0.877) and is below that of the trivial all-negative classifier, which shows why accuracy is uninformative here. In the GBDT model, we force the prediction to positive for every respondent who answered `HAREHAB1` (cardiac rehabilitation): this question is only asked of heart-attack survivors, and all 773 such training respondents are positive. We do not apply this rule to the logistic model, which labels only 154 of the 238 such test respondents as positive. Applying the same rule would raise its F1 to roughly 0.444, still below the GBDT model, so the rule alone does not explain the gap between the two models.

REFERENCES

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